What Does Deferral Buy? Measuring Provisional Identity in Streaming Linkage [Experiment, Analysis & Benchmark]

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ABSTRACT

Streaming linkage can keep uncertain reports retrievable while postponing their final identity assignment. We measure this design with an event-time engine that emits one irrevocable outcome per request, enforces a seven-day deadline, and records complete outputs. On San Diego's May stream, five matched scorer fits give mean pairwise F_1 of 0.415 for arrival commitment, 0.449 for a score band, and 0.467 for a staleness gate. The gate exceeds a ten-day cutoff by 0.0246 (95% fit-seed interval $[0.0093, 0.0400]$). A newly frozen retrospective July evaluation gives a gate–cutoff difference of 0.0180 ($[0.0135, 0.0225]$). Hiding pending identities changes both retrieval and realized waiting. Event-level analyses expose newly recovered and lost positive pairs, different merge denominators, and both arrival- and deadline-initiated claims. Actual candidate graphs separate direct coverage from transitive reachability. The logistic scorer reverses the gate–cutoff ordering on July. The findings concern workflow-derived identity labels; scorer dependence, annotation uncertainty, and cross-city checks bound their operational interpretation.

KEYWORDS

entity resolution, streaming, decision timing, prefix-safety

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The source code, data, and/or other artifacts have been made available at https://github.com/yyting2006-ai/deferral-streaming-linkage.

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1 INTRODUCTION

A municipal service desk may receive several reports about a broken streetlight. Linking them can avoid duplicate work, while an incorrect link can suppress an unrelated problem. If linkage triggers an irreversible action, the system must decide when a tentative identity becomes binding. A record can remain unresolved while still being visible to later reports (Figure 1). This changes what later records retrieve and what evidence is available when the original record is finally assigned.

Immediate output and causal output impose different conditions. An immediate policy binds an identity at arrival. A causal policy can bind it later, using only information available at emission. Delayed decisions and irreversible matching already have a substantial history in online matching and streaming systems [1, 7]. Our question is empirical: within a learned streaming linker, what does a retrievable provisional identity buy, and how much of its apparent benefit can a simple age cutoff reproduce?

The measured system choice. Our pipeline retrieves candidate issues using address and token indexes and scores their representatives. Arrival commitment either creates an issue or merges immediately. A score band instead holds marginal matches; a gate holds only marginal matches to sufficiently old representatives. Each held record forms a singleton provisional identity with a timer. A later merge can finalize that target before attaching to it, or the timer can re-score it. Previously committed identities never change. This exposes a precise state-management choice while retaining a fixed scorer and fixed retrieval budgets.

Retrieval settings do not determine a policy-independent candidate set. A provisional identity may become a candidate for later requests and may change the representatives and capacities they encounter. We therefore record actual candidate snapshots and distinguish end-to-end policy comparisons from a conditional analysis of one frozen candidate trace. The latter cannot recover the adaptive feedback of a different policy.

Evidence from matched comparisons. On May, the score band changes mean F_1 from 0.415 to 0.449, and the gate reaches 0.467. The ten-day cutoff reaches 0.442. The gate–cutoff contrast is 0.0246, with a five-fit interval of $[0.0093, 0.0400]$ (Table 1). May had influenced earlier exploration. We consequently preserve its exploratory

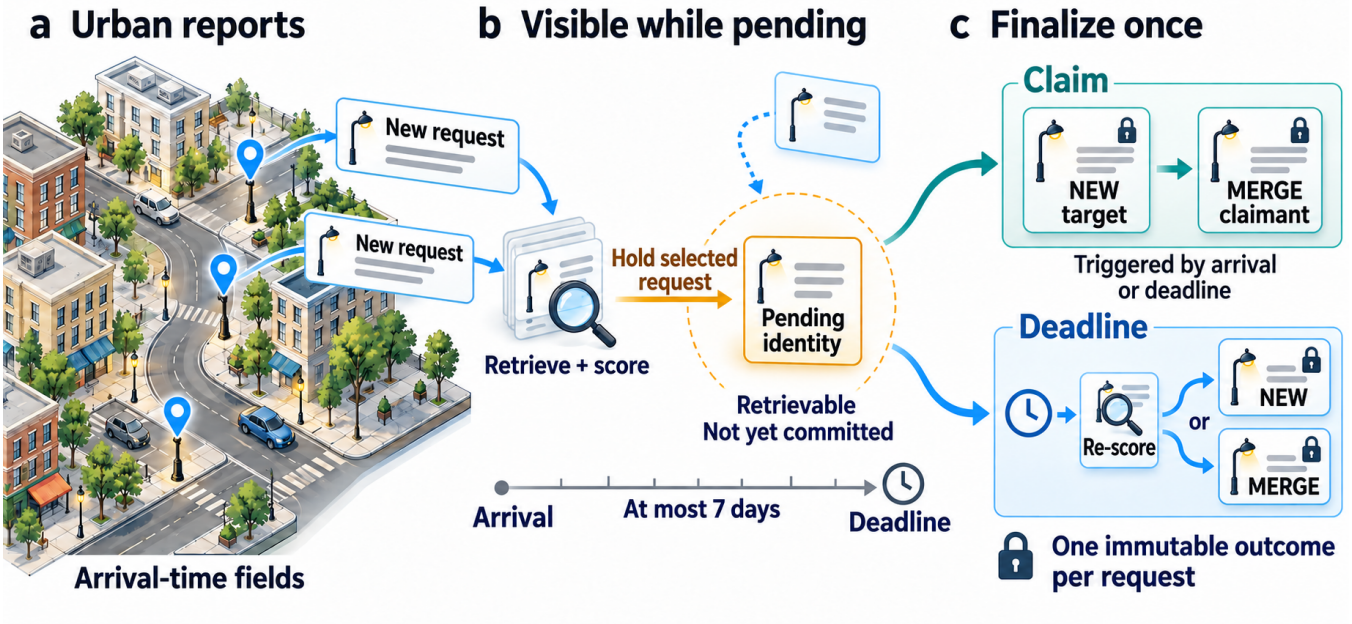


Figure 1: Selected deferred path. One request remains retrievable while pending. A claim emits the target's NEW before the claimant's MERGE and can originate at arrival or deadline. Otherwise, deadline re-scoring emits NEW or MERGE. Each request has one immutable outcome. AI-assisted schematic checked against Algorithm 1; immediate paths omitted.

status and freeze the complete revised protocol before a later July evaluation. There the same gate and cutoff configurations differ by 0.0180 ([0.0135, 0.0225]). These intervals quantify scorer-fit variation on fixed monthly streams, rather than city-level generalization.

The mechanism evidence is more specific than a single aggregate gain. Visibility-off controls change early claim opportunities and actual waiting times as well as retrieval. Correct-pair sets contain both additions and losses, and claims can originate at an arrival or another record's deadline. A higher adjudicable-merge fraction can coexist with fewer adjudicable merges in absolute terms. These distinctions rule out reading a net change as proof that no individual record benefited, or treating an interval spanning zero as an equivalence result.

Contributions. We contribute an explicitly timed, irrevocable streaming-linkage protocol with executable output-prefix checks; matched policy and scorer comparisons with a recorded selection timeline; and event/candidate analyses that separate pair changes, merge-support changes, and finalization triggers. The accompanying artifact preserves inputs, configurations, result rows, and verification code. The reference partition is induced by administrative workflow links. Its closed-world negatives, retrospectively available labels, and the limited human-annotation check constrain the interpretation throughout.

2 RELATED WORK

Our study connects record linkage, online matching with delays, and streaming systems. These literatures already address uncertain matches, delayed decisions, and evolving results. We examine how selective commitment and the visibility of provisional identities interact in a particular streaming linkage pipeline.

Entity resolution and progressive processing. Swoosh formalizes entity resolution through match and merge functions and studies properties that permit efficient resolution algorithms [4]. Progressive ER instead emphasizes the order in which useful matches become available under limited computation. Maciejewski et al. organize this process into filtering, weighting, scheduling, and matching [12]. Their study also discusses progressive processing of streaming data; progressive ER should therefore not be characterized as universally requiring a fixed evidence set. Our experiments hold retrieval settings and the learned scorer constant across internal policy comparisons, while changing the treatment of uncertain arrivals. This evaluates a commitment policy within one pipeline rather than a general alternative to progressive comparison scheduling. Because policies alter index state, holding settings constant does not fix the candidate pairs encountered.

Online matching with delays. Emek, Kutten, and Wattenhofer study minimum-cost perfect matching with delays in a known metric space [7]. Requests arrive online, may wait to be paired with later arrivals, and are matched without preemption. The objective combines matching distance and waiting time, with competitive guarantees relative to an offline optimum. Commitment timing and irreversibility are thus already central to this model. Our setting differs in its learned scores over heterogeneous records, multi-record issue states, and budgeted retrieval from a changing index. We evaluate linkage quality relative to workflow-derived partitions and measure logged merge operations and latency, rather than a metric matching cost. The contribution is an empirical examination of

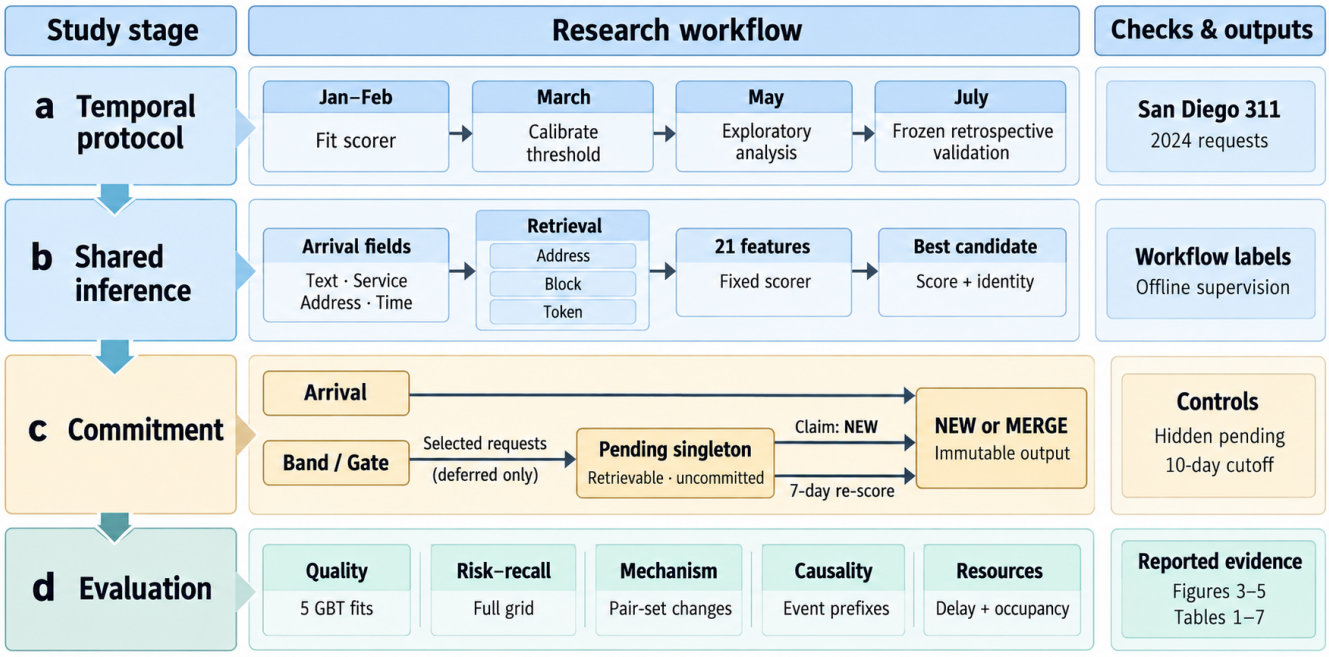


Figure 2: Study design. January–February fits the scorer; March calibrates its threshold by a complete grid. May remains exploratory and July evaluates the newly frozen configuration. Address, block, and token retrieval feed a binding event-time state machine. Workflow labels supervise fitting and evaluation, while future reference labels are never online lookup inputs. AI-assisted schematic checked against the released protocol and implementation.

these system choices, not the introduction of irreversible delayed decisions.

Abstention and deferral. The Fellegi–Sunter framework distinguishes links, non-links, and possible links requiring further review [8]. Reject-option classification trades coverage against error [5, 9]; learning to defer explicitly routes selected decisions to an expert [15]. Abstention alone need not imply delegation to another actor. In our implementation, an uncertain record remains available to the same linking system, which may act on subsequent arrivals or re-score it when its waiting interval expires. The comparison of visible and invisible pending records tests a concrete state-management choice. It does not establish that waiting is generally useful or useless, nor that delayed evidence cannot help individual records.

Streaming completeness and output revision. The Dataflow model separates event-time windows from triggers that determine when results are emitted, and supports different refinement modes, including accumulation and retraction [1]. Watermarks provide progress estimates; heuristic watermarks do not certify that all relevant data have arrived. This literature already exposes correctness–latency trade-offs. Our application contract requires emitted merge decisions to remain binding. The experimental question is how to manage uncertain identity state before emission under that contract. This is distinct from claiming that stream-processing systems cannot express delayed or irreversible outputs.

Candidate generation and external comparisons. Blocking and filtering reduce comparison cost while affecting which direct record pairs reach a matcher [6, 16]. Direct candidate-pair recall and final partition recall are different quantities: transitive merges can recover a pair that was never directly compared. We therefore interpret the reported candidate coverage as a property of the measured retrieval procedure, not as a proved upper bound on every policy's final recall. Splink uses its own blocking and estimation procedures [14], so its comparison is end to end. Our internal policies enforce a true size cap of 20, but this alone does not equalize candidates with an external linker or isolate commitment timing as the source of an end-to-end performance difference.

Risk calibration under distribution shift. Conformal risk control provides risk guarantees under stated conditions [2]; adaptive and nonexchangeable conformal methods address settings beyond ordinary exchangeable calibration [3, 10]. Our threshold calibration and subsequent monthly evaluation are empirical procedures. They do not provide an online guarantee on the false-merge proportion, and temporal drift alone does not establish that all formal calibration methods are inapplicable. Comparisons with threshold policies concern the evaluated settings and their observed risk–coverage trade-offs.

Evaluation and scope. ER and coreference measures assign different weights to record-level, pair-level, and entity-level errors [11, 13]. We report pairwise partition metrics together with decision-level merge errors because the final partition does not describe the sequence of emitted actions. These measures are complementary; record-averaged measures are not generally insensitive to

erroneous merges. Our conclusions concern the observed streams, their labeling procedures, and the tested configurations. Aggregate lag-bin changes and policy differences describe associations; they do not by themselves identify a causal decomposition of the benefit of deferral.

3 PROTOCOL AND MEASUREMENTS

Arrival, commitment, and identity. A request x_i has arrival time a_i, final commitment time c_i, and final issue identity d_i. Time is a UTC timestamp, distinct from the record index i and the score threshold τ. A final output event contains its emission time, request identifier, action, target identity, payload, and sequence number. New creates a stable identity; Merge attaches one request to an already finalized identity. Each observed request emits exactly one such event. An internal pending state is not a final output. A committed request is never moved to another identity, and finalized identities are never merged together.

Definition 1 (Immediate commitment). *An immediate policy satisfies $c_i = a_i$ for every request. Its action and identity use only the input prefix available at a_i and fixed prior state.*

Definition 2 (Bounded causal commitment). *For a fixed maximum delay D, a bounded causal policy satisfies $a_i \le c_i \le a_i + D$. The final identity d_i and its output event use only arrivals no later than c_i and fixed prior information. Conditional on the same initial state, scorer, configuration, and randomness, two streams identical through any time T must emit identical ordered event sequences through T, including event counts, times, objects, actions, targets, and payloads. The identity d_i is binding at c_i. Here D is an elapsed duration; a reported day is exactly 86,400 seconds.*

Immediate commitment is the special case $D = 0$. Nonanticipation alone does not impose immediate commitment: evidence arriving after a_i but no later than c_i may legitimately influence a delayed output. These definitions concern inference with a fixed scorer. They do not establish when retrospective training labels became available.

Workflow reference partition. We construct undirected components from the available official child–parent links. The labeled domain $\mathcal{K}$ contains records participating in these links. For two records in $\mathcal{K}$, equal components are positive and unequal components are negative under a *closed-world workflow assumption*. This assumption does not exclude missing cross-component links. A pair touching an unlabeled record is unknown. Accordingly, "correct" and "wrong" below mean agreement or disagreement with this frozen workflow partition, rather than an independently established operational event identity. Without additional labels, recurrence or operational inconvenience cannot explain a counted error.

Two different units of error. Partition precision, recall, and F_1 use all unordered predicted pairs in $\mathcal{K}$ and all positive reference pairs whose endpoints occur in the replay window. In contrast, a merge decision is the directed pair of the finalized request and the first representative of its chosen destination. Write C, W, and U for correct, wrong, and unknown merge decisions, and $M = C + W + U$. We report

$$\mathrm{FMP}_{\mathrm{dec}} = \frac{W}{C+W}, \qquad \mathrm{coverage}_{\mathrm{dec}} = \frac{C+W}{M}.$$

Both ratios are undefined when their denominators vanish. Coverage is the adjudicable fraction of merges, not the fraction of requests receiving service. For every arm we also count total requests, final events, New, Merge, pending requests, and each of C, W, U. A single merge may induce several partition pairs; the two error units must not be interchanged.

4 AN IRREVOCABLE EVENT-TIME IMPLEMENTATION

Candidates and scores. The actual index has three channels: exact normalized address within service, hundred-block address within service, and token overlap. Each address channel selects a contiguous slice of up to 40 calendar-ordered postings around the query date's insertion point; this is not absolute-distance nearest selection. For an arriving request these are recent entries, whereas deadline rescoring retains the request's original date. The text channel accumulates inverse-document-frequency weights for the eight highest-IDF query tokens, scanning the latest 60 postings per token and retaining up to 60 identities. The deduplicated union can contain up to 140 identities. There is no independently executed dense, alias, or category-retrieval channel. Candidate identities are deduplicated and ordered by their stable identity keys. We score the first representative of each eligible identity. The index is pruned on UTC day changes. At each scoring event, an additional exact elapsed-time check excludes identities whose latest member is more than 60 days old; self-candidates and identities with 20 actual members are also excluded. A cutoff arm additionally excludes representatives older than its specified cutoff at scoring time.

Our boosted classifier receives 21 feature slots describing lexical/address, service, time, and issue state. In the singleton-candidate training replay, issue size is constant at $\log(2)/\log(11)$, the historically named `service_rare` slot is a constant nonempty-service indicator, and both unused address-embedding slots are zero. The artifact lists every feature, its transform, and its observed training range. It does not describe unused embedding slots as an executed dense retrieval method. Score ties use the lexicographically smallest stable identity key, and token-IDF ties use lexical token order. Index postings are added once, when an identity first becomes visible; they are not rebuilt after every attached member.

Policies. Let s^* be the best candidate score. Arrival commitment merges when $s^* \ge \tau$ and otherwise creates a new identity. The band policy holds a request when $D > 0$ and $\tau \le s^* < \tau + \delta$. The gate holds it only when this band condition also holds and the best representative is at least h elapsed days old. Otherwise it follows arrival commitment. A pending identity contains exactly one uncommitted request. Its timer is scheduled for $a_i + D$; by default it is immediately retrievable. Visibility-off policies retain the same hold predicate but hide this identity until it is finalized as New.

Claims and timers. A later request claims a pending target only when its own policy selects an actual merge. The target first emits New, then the claimant emits Merge into that now-final identity; the target timer is canceled. If the later request is itself held, it creates a separate singleton pending identity. A deadline resolution can likewise claim another pending identity. No path carries previously

Algorithm 1 Bounded causal band or gate policy

Require: Fixed scorer, τ, δ, D, h, visibility v, capacity $B \geq 1$

procedure FinalizeNew(p, t, g)
  Assert p is a singleton with an uncommitted request
  Cancel its timer; remove pending status; ensure p is visible
  Emit final New for its request at t, trigger g
end procedure
procedure Attach(x, p, t, s, g)
  Assert x is uncommitted, $x \notin p$, and $|p| + 1 \leq B$
  if p is pending **then**
    FinalizeNew(p, t, claim)
  end if
  Cancel and retire x's pending singleton, if present
  Attach only x; retain the latest member date if newer
  Emit final Merge(x, p) at t, score s, trigger g
end procedure
procedure Arrive(x, t)
  Process all timers due at or before t
  $(p, s) \leftarrow$ BestEligible(x, t)
  if $p = \emptyset$ or $s < \tau$ **then**
    Create stable singleton p_x; FinalizeNew(p_x, t, arrival)
  else if $D > 0$, $\tau \leq s < \tau + \delta$, and (band or age$(p, t) \geq h$) **then**
    Create singleton pending p_x; schedule $t + D$
    if v **then**
      Index p_x
    end if
  else
    Attach(x, p, t, s, arrival)
  end if
end procedure
procedure Deadline(x, t)
  if x is already final **then**
    return
  end if
  $p_x \leftarrow x$'s pending singleton
  $(p, s) \leftarrow$ BestEligible(x, t), excluding its own identity
  if $p \neq \emptyset$ and $s \geq \tau$ **then**
    Attach(x, p, t, s, deadline)
  else
    FinalizeNew(p_x, t, deadline)
  end if
end procedure
procedure Finish
  Process all remaining timers at their individual exact due times
end procedure

committed members into a different issue. Timers at a timestamp execute before arrivals at that timestamp, with stable identity order for tied timers and stable record order for tied arrivals. Identities use the key `I:` followed by their anchoring request identifier.

Lemma 1 (Finality and bounded causal output). *Given a fixed deterministic scorer, configuration, and initial state, and processing arrivals and timers in the stated order, Algorithm 1 emits one final event per observed request, never changes its final identity, and satisfies bounded causal commitment when the event clock is advanced through each deadline, including the final drain.*

Proof. Each transition reads only the current index and already-arrived records. Identical input prefixes therefore give identical state transitions, timer firings, and ordered outputs by induction. A request emits immediately, on a claim after its arrival, or at its scheduled deadline. Timers are processed at their exact due times, so no request remains unresolved after $a_i + D$. Every finalization removes the pending state, and stale timer entries are ignored. The attach operation accepts only an uncommitted source and first finalizes a pending target, so a final identity never changes and a request cannot emit twice. The capacity test counts actual members before attachment. □

This argument applies to the stated algorithm; executable tests and the real-stream output audit below check the released implementation. At month end we drain remaining timers at their exact due times using only records supplied in that month's replay. No later-month requests are added during this drain. We measure latency as $(c_i - a_i)/86400$ in days and separately summarize all records and held records. Entry into pending, claim finalization, and deadline resolution are distinct counters. Resource measurements include peak pending state and processing time under the reported local CPU conditions.

5 EXPERIMENTAL DESIGN

Data and retrospective labels. San Diego's 2024 closed-request snapshot contains 429,924 requests and 57,463 parent-link fields, of which 56,153 resolve within the snapshot. About 20.3% of requests participate in the linked domain. The closed-request sampling frame can underrepresent unresolved cases, and a final export does not prove that every description or address field was unchanged after arrival. Scoring uses only the designated request fields, excluding closure/status fields. We build the reference components from the complete snapshot, including links that may have become available after the training period. Link-creation times are unavailable. Thus this is retrospective weak supervision; chronological feature replay and a fixed-model causal audit do not certify historical label availability or a faithful prospective deployment.

We select monthly windows by the source's local calendar, interpret its naive timestamps as America/Los_Angeles, and convert them to UTC before replay. Training and scoring use UTC calendar-day features; timers use exact elapsed seconds, including across daylight-saving changes. Each monthly replay starts with an empty index. The ordering key is UTC arrival time followed by the source request identifier. The artifact records this convention and the raw-file checksum.

Training. January–February supplies 71,923 records. A fixed candidate replay uses the first 22,000, retains each weak negative candidate with probability 0.35 using sampling seed 13, and produces 441,455 training pairs, including 3,080 positives. Equal known components are positive; all other sampled pairs, including unknown ones, are treated as negative for fitting. This noisy-label convention is distinct from evaluation, which leaves pairs outside the known domain unadjudicated. All policies share these training pairs. The

boosted scorer uses 160 trees, learning rate 0.1, depth 3, subsample 0.5, and minimum leaf size 20. We refit it with seeds $\{1, 2, 3, 4, 13\}$. The second family is standardized logistic regression with $C = 1$, LBFGS, maximum 4,000 iterations, and seed 13.

Nonmonotonic calibration. For each family we run the arrival policy on March with seed 13 at thresholds .01, .02, .0358, .05, .07, .1, .15, .2, .3, .5, .7, .9, and .99. A point is volume-eligible only if it merges at least 1% of requests and has at least 100 adjudicable merges. Among volume-eligible points with decision FMP at most 0.40, we maximize positive-pair recall, breaking ties by higher F_1 and then higher τ. If none is feasible, the rule selects the minimum observed FMP among volume-eligible points, breaking ties by higher recall, F_1, and then τ, and reports infeasibility. This finite-grid rule assumes no monotonic relationship: changing a threshold changes later identity states. Zero-merge runs are not successful calibration solutions. The 0.40 budget is a chosen workflow-proxy constraint, not a production safety guarantee.

Selection record and evaluation windows. Historical band, gate, and cutoff choices were influenced by already-viewed experiments. In particular, the old April cutoff sweep had five seeds for $X = 5, 10$ but only three for $X = 14$; its selection script excluded the incomplete candidate. It did not establish a complete April argmax at $X = 10$. Original first-view and freeze times cannot be recovered and remain marked unknown in the released stage manifest. No file modification time is presented as a preregistration timestamp.

For the repaired implementation we froze the full experiment protocol before viewing its outcomes on 27 October 2026. We retain $\delta = .5$, $D = 7$ days, $h = 5$ days, and cutoff $X = 10$ from historical exploration. March calibrates the new model-specific threshold; that choice is persisted before any revised May/July quality run. May remains exploratory. A metadata search found no earlier July quality outputs, so July supplies a newly frozen retrospective validation of the gate–cutoff comparison. It is not prospective deployment or a reconstruction of an earlier preregistration. All five fitted scorers use the same family-specific threshold and the same policy settings.

Controls and uncertainty. For May and July we replay arrival, band, gate, cutoff, invisible band, and invisible gate with all five shared seeds. The logistic check includes arrival, band, gate, and the same ten-day cutoff in both months. We retain the entire 13-point May boosted threshold curve, one-seed April parameter grids, and $D = 3, 14$ sensitivity runs. The main $D = 7$ rows complete that delay comparison. These secondary runs do not select the primary July comparison. Means and sample SDs describe five scorer fits on a fixed stream. Paired differences use exactly the same five seeds, with 95% t intervals (four degrees of freedom). Intervals are pointwise and unadjusted for multiple contrasts. They measure fit-seed variability, not uncertainty over cities, label acquisition, or the historical policy-selection process.

Version separation. The historical engine could move previously committed members and processed nominal deadlines only at later arrivals or at window termination. It also double-counted the first member in its issue-size counter. We therefore retrain all scorers after fixing the counter and report the repaired implementation as a separate experiment series. Primary tables use only this series. Historical transfer and external-linker results, where retained, are explicitly labeled and do not serve as measurements of the corrected state machine.

6 RESULTS OF THE CORRECTED REPLAY

6.1 Matched quality and scorer dependence

The gate improves on the ten-day cutoff in every matched boosted fit in both months (Table 1 and Fig. 3). Its mean difference is 0.0246 on May, or 2.46 percentage points of F_1, and 0.0180 on July, or 1.80 points. The paired 95% intervals are $[0.0093, 0.0400]$ and $[0.0135, 0.0225]$. The band–cutoff intervals include zero: May 0.0067 $[-0.0128, 0.0261]$ and July 0.0045 $[-0.0002, 0.0093]$. Thus the stronger evidence for a residual over this cutoff concerns the gate.

On May, hidden Band minus Arrival is 0.0007, with interval $[-0.0132, 0.0145]$. Hidden Gate minus Arrival is 0.0101, with interval $[-0.0024, 0.0227]$. These are imprecise contrasts, not equivalence tests. Visible minus hidden effects are 0.0336 $[0.0185, 0.0488]$ for Band and 0.0422 $[0.0323, 0.0520]$ for Gate. Visibility alters both candidates and claim times, so these differences do not isolate retrieval from waiting.

On July, hidden Gate minus Arrival is 0.0058 with paired 95% interval $[0.0036, 0.0081]$; all five fit differences are positive. Visible Gate adds 0.0266 $[0.0212, 0.0321]$ over hidden Gate. The hidden contrast also changes selection, deadline re-scoring, and later state, so it is not a pure waiting effect.

Scorer dependence is substantial (Table 2). With logistic scores, Band lowers F_1 in both months. Gate exceeds Cutoff on May by 0.0253, but falls below it on July by 0.0077. A common numerical band width need not represent the same uncertainty across differently calibrated scorers. These single-fit checks bound generality; the boosted ordering is not a scorer-independent property of deferred commitment.

6.2 Calibration and threshold alternatives

The March rule selects $\tau = .0358$ for boosting (decision FMP 0.3986, recall 0.4887) and .20 for logistic regression (FMP 0.3891, recall 0.4454). The selected boosted threshold exceeds the budget on May (FMP 0.4362). In the complete May grid, .15 satisfies the volume rule and has FMP 0.3979, recall 0.3944, and $F_1 = 0.4147$. Thus threshold adjustment is a viable observed alternative; no impossibility claim follows from this grid. FMP is nonmonotonic: raising τ from .5 to .7 changes May FMP from 0.3287 to 0.3650. At .99, only four May merges remain, one adjudicable and wrong; it fails the volume rule. March's logistic .99 point makes no merges and has undefined decision FMP.

6.3 Merge support and pair-level changes

Table 3 shows why decision FMP needs its denominators. On May, Band increases adjudicable-merge coverage from 28.73% to 32.74% while reducing the absolute adjudicable count from 2,391 to 2,248; Gate has 2,358. On the common sources adjudicable under both policies, May FMP changes from 0.3911 to 0.3122 for Band ($n = 1,874$), and from 0.3929 to 0.3380 for Gate ($n = 2,092$). July gives 0.3907 to 0.2872 ($n = 1,894$) and 0.4034 to 0.3472 ($n = 2,169$), respectively. A pooled service-by-target-age standardization over shared strata also retains negative May differences (−0.1152 and

Table 1: Primary event-time replay. Every row uses the same five scorer seeds $\{1, 2, 3, 4, 13\}$, a March-selected threshold of .0358, and a true capacity of 20. Precision/recall evaluate the known workflow domain; decision FMP and coverage use the merge denominators defined in Section 3. May is exploratory; July follows the newly frozen protocol.

Window	Policy	F_1 (mean±SD)	Precision	Recall	Decision FMP	Coverage
May	Arrival	0.415 ± 0.003	0.395	0.437	0.439	0.293
May	Band	0.449 ± 0.016	0.526	0.391	0.309	0.344
May	Gate	0.467 ± 0.014	0.510	0.431	0.340	0.309
May	Cutoff (10 d)	0.442 ± 0.003	0.473	0.416	0.378	0.320
May	Hidden band	0.415 ± 0.010	0.410	0.421	0.411	0.315
May	Hidden gate	0.425 ± 0.011	0.415	0.435	0.421	0.296
July	Arrival	0.469 ± 0.007	0.419	0.532	0.428	0.304
July	Band	0.488 ± 0.007	0.591	0.416	0.290	0.344
July	Gate	0.501 ± 0.007	0.551	0.460	0.340	0.311
July	Cutoff (10 d)	0.483 ± 0.006	0.514	0.457	0.358	0.323
July	Hidden band	0.473 ± 0.009	0.439	0.512	0.406	0.331
July	Hidden gate	0.475 ± 0.007	0.440	0.515	0.405	0.304

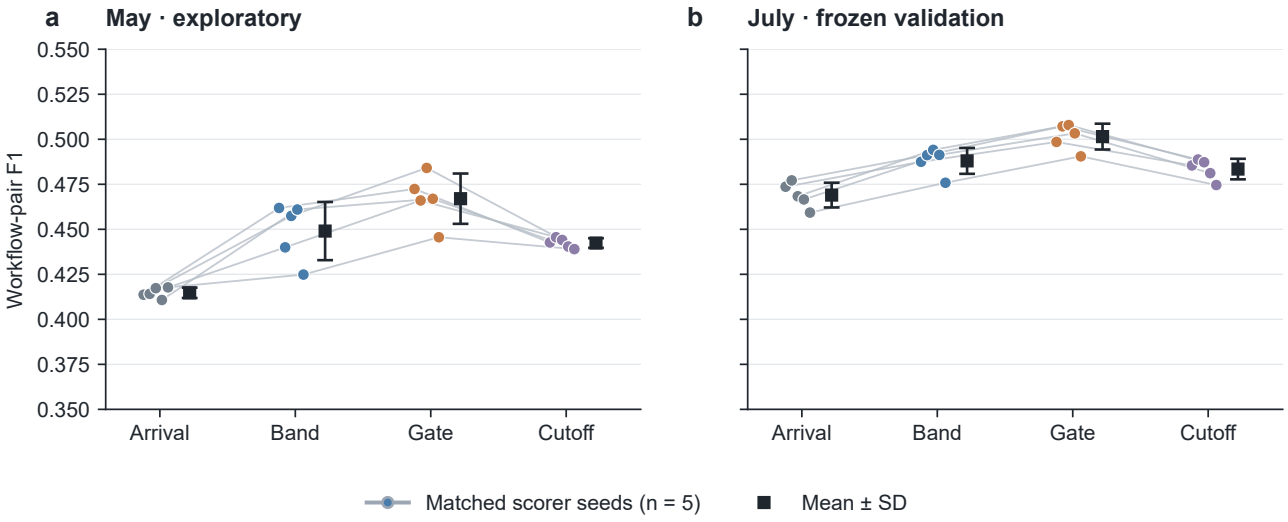


Figure 3: Paired policy comparisons. Colored points show all five GBT fits; lines connect the same fit. Black squares and whiskers show mean and sample SD. Variability concerns scorer fitting on each fixed workflow-labeled stream. May is exploratory; July evaluates the newly frozen configuration.

Table 2: Logistic scorer, seed 13, separately March-calibrated $\tau = .20$. The gate and the same ten-day cutoff are both measured; these are single-fit descriptive comparisons.

Policy	May F_1	FMP	July F_1	FMP
Arrival	0.400	0.419	0.415	0.428
Band	0.299	0.328	0.313	0.314
Gate	0.432	0.366	0.411	0.383
Cutoff (10 d)	0.407	0.403	0.419	0.406

−0.0469). Conditioning does not recover unknown truth or hold the selected target fixed.

Both strategies recover and lose positive pairs (Figure 5). May Band gains 512 and loses 1,011, while Gate gains 506 and loses 673. July gains are 352/281 for Band/Gate, against losses of 1,095/742. Net positive-pair recall therefore falls for these runs. The gains in F_1 accompany a precision–recall tradeoff, not a uniform recovery of additional matches. The corresponding avoided/new wrong directed operations are 765/442 and 587/354 on May, and 874/419 and 626/361 on July. The artifact reports common, gained, lost, and neither-positive sets for every lag bin, with separate target-age bins for wrong operations.

The trigger audit further limits an arrival-evidence explanation. May Band finalizes 1,316 pending targets by claim: 662 claims originate at a new arrival and 654 at another request's deadline. Gate's

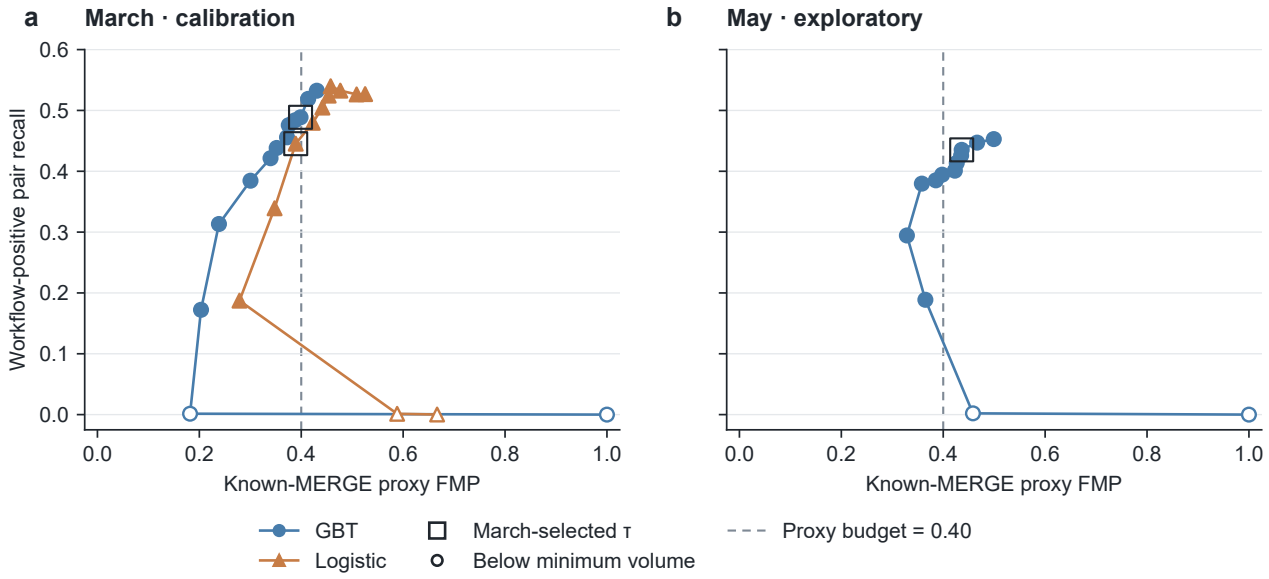


Figure 4: Complete arrival-policy threshold curves, seed 13. Filled points satisfy the minimum-volume rule; open points do not. Square outlines identify the March-selected threshold, including its May operating point. The dashed line marks the 0.40 known-merge proxy FMP budget. Connections follow threshold order without imposing monotonicity. Undefined-risk points remain in Source Data and are omitted from the plane.

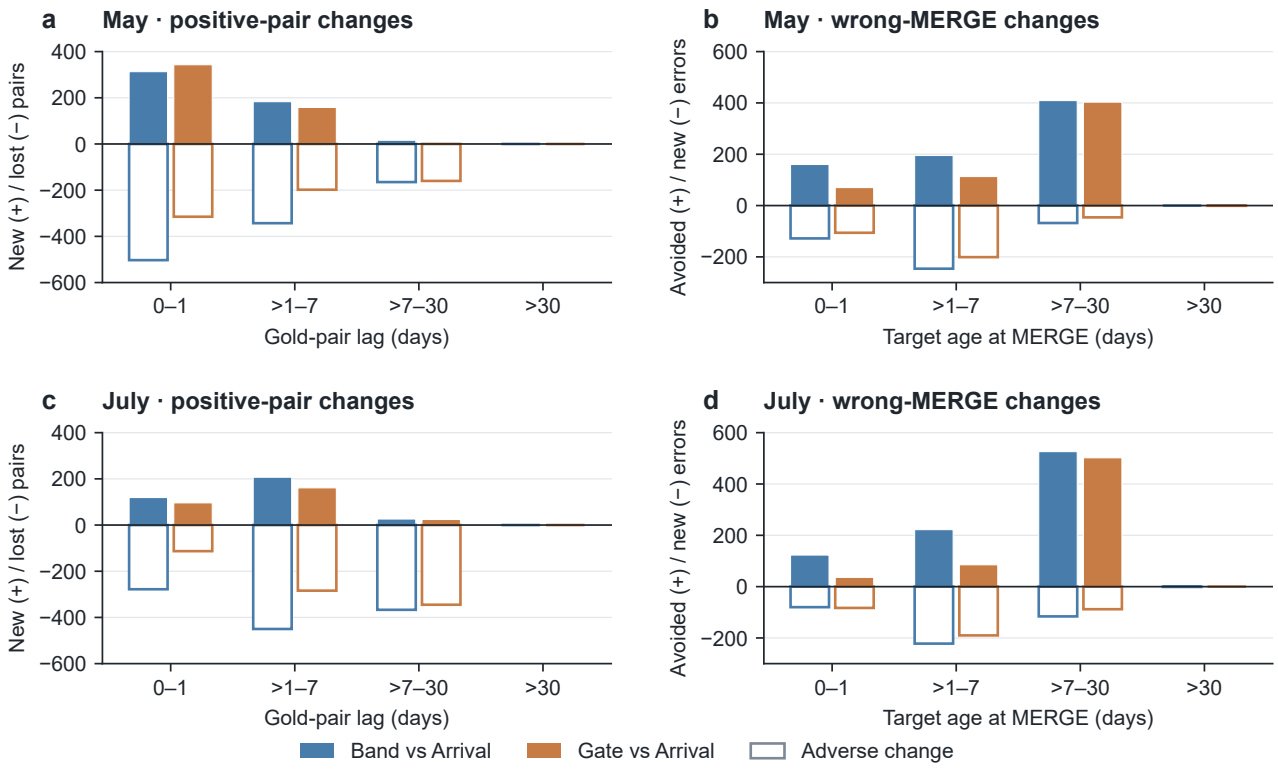
Table 3: Complete merge denominators for seed 13. Each May arm finalizes all 35,915 requests and each July arm all 37,372, exactly once; NEW + MERGE equals the request count and final pending count is zero. Known merges = Correct + Wrong. Every outcome is relative to the closed-world workflow partition.

Window	Policy	NEW	MERGE	Correct	Wrong	Unknown	Coverage
May	Arrival	27,593	8,322	1,348	1,043	5,931	0.287
May	Band	29,048	6,867	1,528	720	4,619	0.327
May	Gate	28,100	7,815	1,548	810	5,457	0.302
May	Cutoff (10 d)	28,582	7,333	1,397	898	5,038	0.313
May	Hidden band	28,268	7,647	1,329	997	5,321	0.304
May	Hidden gate	27,686	8,229	1,377	1,014	5,838	0.291
July	Arrival	29,179	8,193	1,412	1,104	5,677	0.307
July	Band	30,674	6,698	1,559	649	4,490	0.330
July	Gate	29,734	7,638	1,538	839	5,261	0.311
July	Cutoff (10 d)	30,177	7,195	1,453	860	4,882	0.321
July	Hidden band	29,861	7,511	1,419	1,008	5,084	0.323
July	Hidden gate	29,297	8,075	1,449	1,022	5,604	0.306

611 claims split 542/69. Among Band's 512 newly recovered positive pairs, the later endpoint's operation is arrival-claim for 58, deadline-claim for 26, its own deadline for 50, and immediate output for 378. Gate's 506 split 36/2/14/454. These event associations describe the realized execution; they do not identify a causal mediation effect of new evidence.

6.4 Candidates: direct coverage and relaxed bounds

Actual scored query–representative pairs cover only a fraction of reference positive pairs (Table 4). This fraction is not a ceiling on partition recall, because transitive attachment can recover unscored pairs. The union of actual representative edges gives an undirected



Seed 13; descriptive set differences. Unchanged pairs are included in Source Data.

Figure 5: Exact set changes relative to Arrival (GBT seed 13). (a,c) Newly recovered and lost positive pairs, grouped by gold-pair lag. (b,d) Avoided and new directed wrong merges. Avoided operations use Arrival's target age; introduced operations use the deferral policy's target age. Rows show May and July. Open bars denote adverse changes; bins are $[0, 1]$, $(1, 7]$, $(7, 30]$, and $(30, \infty)$ days. Counts are dependent descriptive set differences under the workflow proxy. Unchanged pairs and trigger decompositions are included in Source Data.

graph. Within each connected component, a gold-aware construction of pure connected groups of at most 20 gives a feasible static lower bound. A combinatorial upper bound distributes each gold label into groups of size 20 while relaxing connectivity within those groups. The fully uncapped reachability bound is 1.0 in both months. Neither bound is an exact online oracle; both ignore arrival order. They are conditional on each policy's realized graph, not on all possible retrieval histories.

Freezing Arrival's recorded candidate rows gives a separate diagnostic. Static threshold selection yields May $F_1 = 0.4177$; adding the ten-day age filter to those same rows gives 0.4003. This differs from the adaptive cutoff replay, which can change later representatives and candidates. The frozen analysis cannot implement pending states, later retrieval, or a waiting counterfactual. Historical external-linker runs use their own blocking and likewise cannot isolate commitment on a shared candidate population.

6.5 Actual delay and resource costs

Every core replay emits one final event per request and ends with zero pending requests. At $D = 7$, no held record exceeds seven elapsed days, including the end-of-month drain (Table 5). Hiding pending identities changes actual latency: all hidden records reach their own deadline, while visible targets can finalize on a claim. May's mean pending occupancy over the full replay horizon is 913.7 for Band and 251.9 for Gate, versus 1,145.2 and 475.2 for their hidden counterparts. This includes the final timer drain.

The one-fit May delay sensitivity gives Band F_1 of 0.4304 at $D = 3$ and 0.4203 at $D = 14$; Gate gives 0.4462 and 0.4437. Held median, p95, and maximum equal the chosen deadline in all four runs, with zero overruns. Longer permitted waiting is not uniformly beneficial. April's one-fit sensitivity is also nonmonotonic: band widths .3, .5, .7 give $F_1 = .4821, .4753, .4169$. Cutoffs 5/10/14 days give .4208, .4519, .4496. These descriptive grids do not reselect the frozen July settings.

Table 6 reports the same trace-enabled May runs. The all-request mean uses 35,915 requests; held means use 5,740 Band and 1,867

Table 4: Candidate diagnostics, seed 13. All three measures are recall fractions (0–1). Direct coverage uses actually scored positive pairs. Static lower/upper bounds concern the best gold-aware partition of the realized representative graph with size cap 20; they relax online timing. The uncapped transitive upper bound is 1.0 throughout.

Window	Policy	Direct recall	Lower recall	Upper recall
May	Arrival	0.239	0.378	0.907
May	Band	0.338	0.526	0.907
May	Gate	0.290	0.499	0.907
May	Cutoff (10 d)	0.240	0.374	0.907
May	Hidden band	0.263	0.420	0.907
May	Hidden gate	0.248	0.396	0.907
July	Arrival	0.290	0.448	0.997
July	Band	0.429	0.628	0.997
July	Gate	0.352	0.530	0.997
July	Cutoff (10 d)	0.288	0.422	0.997
July	Hidden band	0.320	0.489	0.997
July	Hidden gate	0.303	0.465	0.997

Algorithm 2 Complete event-prefix comparison

Require: Stream S, cutoff T, fixed model/state/randomness ω
1: $S' \leftarrow$ preserve every arrival through T; perturb later content
2: $L \leftarrow \textsc{Replay}(S, \omega)$; $L' \leftarrow \textsc{Replay}(S', \omega)$
3: $P \leftarrow [e \in L : e.\text{time} \le T]$
4: $P' \leftarrow [e \in L' : e.\text{time} \le T]$
5: Assert $|P| = |P'|$
6: **for** $j = 1, \ldots, |P|$ **do**
7: Assert every field of P_j equals the corresponding field of P'_j
8: **end for**

Gate requests ("–" denotes an empty held set). Peak pending is a request count, not memory usage. Wall and process CPU time include candidate-snapshot construction, serialization/writes, final close/flush, and the end-of-window timer drain; they exclude data/model loading, fitting, and post-replay metrics/output persistence. Trace time measures serialization and write calls, excluding the final close/flush. The six-worker shared CPU setting does not measure isolated service capacity.

7 OUTPUT-PREFIX AUDITING

For a fixed model, initial state, and randomness, we compare the full emitted event sequences of an original and a suffix-perturbed stream. At a real timestamp T, both sequences are independently filtered by emission time $\le T$; then length, order, time, request, action, target, and payload must agree. Using the original prefix length to truncate the perturbed run would miss extra early emissions. The executable comparator tests extra, missing, and prematurely emitted events as well as altered ordering, targets, and randomness.

The real-stream check uses 8,153 May 1–7 requests and three cutoffs preserving 2,040, 4,077, and 6,118 arrivals, respectively. With the gate policy, $\tau = .0358$, and $D = 7$, all three normal output-prefix comparisons match. An intentionally invalid positive control uses the complete stream's time-weighted description length to alter actual pair scoring before replay. All three controls change full output payloads. At the first two cuts the differences are score fields only; the third changes one emitted decision. This demonstrates sensitivity to this particular future-information fault, not completeness against every possible leakage pathway. The overlapping cutoffs are diagnostic checks, not three independent statistical samples.

A targeted post-review extension uses all 35,915 May requests, the same fixed gate scorer, and a cutoff of May 20 at 07:00 UTC (midnight local time). It preserves 21,683 arrivals and perturbs the subsequent 14,232. Before the cutoff, the 21,328 outputs comprise 20,961 arrival, 229 claim, and 138 deadline finalizations; claim counts the pending target's NEW. All 138 own-deadline re-scorings occur at exactly seven days and before the month-end drain. The complete output prefixes match field for field after suffix perturbation. At 96 actual equal-time deadline/arrival timestamps, all 98 deadline calls precede arrivals. The instrumented baseline also matches the prior complete May event log. This extends observed deadline coverage, not the scope of the causal guarantee.

Two separate processes set `PYTHONHASHSEED` to 0 and 1, respectively, and replay the same complete week with the band policy and fixed scorer. Both emit 8,153 events and produce identical SHA-256 hashes of their complete serialized event logs. Stable keys, rather than process object identity, determine ties. The audit configuration was fixed separately before main comparisons; its boosted threshold happens to agree with the March-selected primary value. Full models, source versions, and configuration hashes are retained for reproduction.

8 TRANSFER AND ANNOTATION BOUNDARIES

Chicago. The historical Chicago check uses an earlier engine and a six-field export without free text. Among linked reports the median lag is 0.42 days and 84% fall within seven days. In its first March week the largest observed band change among the tested settings is +0.0038 F_1; the selected setting changes the following week's F_1 by −0.0078 in one fitted run. This descriptive contrast is not an estimate of a zero effect and does not validate the repaired gate policy in a second city. Its code, inputs, and original results remain versioned separately from the corrected San Diego experiments.

Louisville annotations. The human-label check begins with 327 records across two weekly waves. The written instructions require separate A/B issue clustering, followed by a third adjudicator. They define an issue as a continuing problem or handling cycle on the same object: matching category or address alone is insufficient; an already resolved problem followed by a new failure should be separate. Annotators are instructed to avoid the other annotator's decisions, model scores, and administrative identity information. Returned-file hashes match the recorded adjudication inputs and final labels, but file consistency alone cannot authenticate the execution of the prescribed independence.

A/B provisional clusterings agree on 90 positive pairs, disagree on 286, and agree on 52,925 negative pairs. Positive-pair Dice agreement is 0.386 and adjusted Rand index is 0.384. Excluding any record flagged for adjudication leaves 258 records, with Dice 0.487

Table 5: Actual finalization timing at $D = 7$ (seed 13); all latency columns are elapsed days. Claim counts the pending target finalized as NEW, whereas Deadline counts its own timer resolution. Held = Claim + Deadline; none exceeds seven days. A claim may be initiated by a fresh arrival or by another deadline; the operation-level decomposition is reported separately.

Window	Policy	Held	Claim	Deadline	All mean	Held mean	Held p95	Maximum
May	Band	5,740	1,316	4,424	0.97	6.04	7.00	7.00
May	Gate	1,867	611	1,256	0.27	5.12	7.00	7.00
May	Hidden band	6,208	0	6,208	1.21	7.00	7.00	7.00
May	Hidden gate	2,575	0	2,575	0.50	7.00	7.00	7.00
July	Band	5,670	1,350	4,320	0.91	5.99	7.00	7.00
July	Gate	1,869	619	1,250	0.26	5.11	7.00	7.00
July	Hidden band	6,131	0	6,131	1.15	7.00	7.00	7.00
July	Hidden gate	2,561	0	2,561	0.48	7.00	7.00	7.00

Table 6: Replay resources on May, GBT seed 13. Mean elapsed waiting is in days; wall, process CPU, and trace-write time are in seconds. All four arms use candidate tracing and the same six-worker CPU setup.

Policy	Mean wait (d)		Peak	Wall	CPU	Trace
	All	Held	pending	(s)	(s)	(s)
Arrival	0.00	–	0	100.12	99.08	6.49
Band	0.97	6.04	1,259	113.80	112.64	7.20
Gate	0.27	5.12	509	104.66	103.53	6.73
Cutoff	0.00	–	0	85.64	84.70	6.71

Table 7: Louisville's short-window check. Intervals are 2.5–97.5% quantiles of 4,000 issue-resampling draws, conditional on the selected candidate set; they are not independent-pair population confidence intervals.

Pair lag (days)	Same / eligible	Rate (%)	Interval (%)
0–1	43 / 537	8.01	3.88–13.64
2–3	18 / 366	4.92	1.56–8.02
4–6	35 / 398	8.79	2.01–16.78
7–13	38 / 639	5.95	1.10–10.58

and adjusted Rand index 0.486; this is a selected-confidence sensitivity rather than a substitute reliability estimate. Adjudication retains 292 records in 226 issues, including 26 multi-record issues, and leaves 35 unadjudicable. All 326 asserted same/different review judgments agree with the resulting partition. Unadjudicable records are excluded from the lag analysis rather than treated as singleton negatives.

Among same-service pairs passing the recorded token-overlap rule, the four lag-bin same-issue rates are nonmonotonic (Table 7). To retain within-issue dependence, we resample final issues with replacement. A within-issue pair receives its issue's multiplicity; a between-issue pair receives the product of the two multiplicities. The resulting sensitivity intervals are broad. Only two waves are available, and all 639 pairs in the last bin cross waves; residual wave and service dependence remains. The sample spans only 13 days. It establishes neither flatness nor a direct test of a mechanism extending through 8–30 days. Released anonymous cluster and candidate-pair relationships reproduce the agreement and resampling calculations conditional on the recorded candidate set.

9 DISCUSSION AND VALIDITY

What is identified. Visibility changes retrieval, claim opportunities, and the state encountered by later arrivals. Equal nominal deadlines do not imply equal realized waiting. Neither a net recall decrease nor an interval containing zero establishes that later evidence never helps. The residual over a cutoff is a complete policy contrast, not an additive decomposition of retrieval and waiting.

Reference labels and support. The workflow partition can omit cross-component matches and encode administrative practices. Decision FMP changes support when policies change whom they merge. The denominator and common-support analyses do not repair unknown labels or randomize support. Closed-request sampling and retrospective annual labels further limit selection and label availability.

Scope of generalization. Five fits quantify fitting variability on fixed streams. July does not remove past policy exploration or establish city-wide transportability. The logistic check has one fit; Chicago is historical; Louisville is selected and adjudication-dependent. Multi-city evaluation and label-acquisition histories would be needed for stronger operational claims.

Candidate and tool comparisons. Frozen-trace filtering and graph bounds complement the adaptive replay. The historical Splink runs have their own blocking and estimation. They provide context, not shared-candidate evidence of a commitment effect. Their original environment was not recorded. Our API check used Splink 5.0.0, which supports term frequency adjustments; the old configurations did not enable them. Differences between these systems cannot be attributed solely to commitment.

Computation and reproducibility. Exact timers and irreversible outputs retain a latency cost. Six-worker CPU timing, including candidate-snapshot serialization in traced runs, does not establish isolated single-core service capacity.

10 ARTIFACT AND REPRODUCIBILITY

The public repository is https://github.com/yyting2006-ai/deferral-streaming-linkage. Release v0.2.0 contains the corrected engine,

complete policy and calibration configuration, individual result rows, paired analysis, candidate-bound code, event-prefix comparator, tests, figure sources, and checksums. Raw event logs can be regenerated from the released San Diego snapshot and frozen scorer recipe; the snapshot's checksum is checked before replay. The Chicago snapshot supports the separately versioned historical check. The Louisville derivative contains fresh anonymous identifiers, cluster relationships, pending flags, wave membership, and precomputed eligible-pair lag bins. It reproduces agreement and issue-resampling calculations but does not independently reconstruct the original text-based eligibility screening or authenticate annotation conduct. Release v0.2.2 adds the post-deadline audit, resource summary, and current manuscript figure mapping.

The earlier engine and its results remain under v0.1.0. Reproduction instructions separate these versions, use a fresh output directory, and reject cache reuse when configurations or provenance differ. Undefined metrics remain undefined in the analysis rather than being accepted as zero-risk operating points. Preparation, calibration selection, run schedules, and original unknown selection times are recorded explicitly. The release is a reproducibility package, not a claim that the venue has certified its artifacts.

11 CONCLUSION

Retrievable provisional identities improve boosted-scorer F_1 on the two measured streams, and the staleness gate exceeds a ten-day cutoff in matched fits. The gain combines fewer wrong merges with losses as well as recoveries of positive pairs. Visibility changes actual waiting and future candidates; it is not an isolated retrieval intervention. The logistic reversal, partial workflow labels, and limited human-label transfer check restrict the claim. An explicit event-time contract and complete output audit make these policy comparisons reproducible while exposing their latency and support costs.

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